

**B.Tech. Degree VI Semester Examination April 2014****ME 606 CAD/CAM**  
(2006 Scheme)

Time : 3 Hours

Maximum Marks : 100

**PART A**  
(Answer *ALL* questions)

(8 × 5 = 40)

- I. (a) Illustrate using a flow chart, the application of computer in the design process.  
 (b) Name three types of automation and compare them with the help of a graph.  
 (c) Compare job shop production and batch production with examples.  
 (d) List four advantages of NC technology.  
 (e) Explain the working of an automatic tool changer with neat sketches.  
 (f) Mention the types of static and dynamic errors in NC machines.  
 (g) Differentiate accuracy and repeatability as applied to NC machines.  
 (h) Illustrate degrees of freedom of a spherical robot with a neat sketch.

**PART B**

(4 × 15 = 60)

- II. Explain with sketches the following: (15)  
 (i) Wire frame modelling  
 (ii) Solid modelling  
 (iii) Surface modelling

**OR**

- III. A 10 station transfer machine is under consideration to produce a component used in a pump. The item is currently produced by more conventional means, but demand for the item cannot be met. The manufacturing engineering department has estimated that the ideal cycle time will be  $T_c = 1$  min. From similar transfer lines, it is estimated that breakdowns of all types will occur with a frequency of  $F = 0.10$  breakdown/cycle and that the average downtime per line stop will be 6 min. The scrap rate for the current conventional processing method is 5%. The casting for the component costs \$ 1.50 each and it will cost \$ 1 per minute to operate the transfer line. Cutting tools are estimated to cost \$ 0.15 per work part. Compute the following measures of line performance. (15)
- (i) Production rate  
 (ii) No. of hours required to meet a demand of 1500 units/week.  
 (iii) Line efficiency  
 (iv) Cost per unit produced

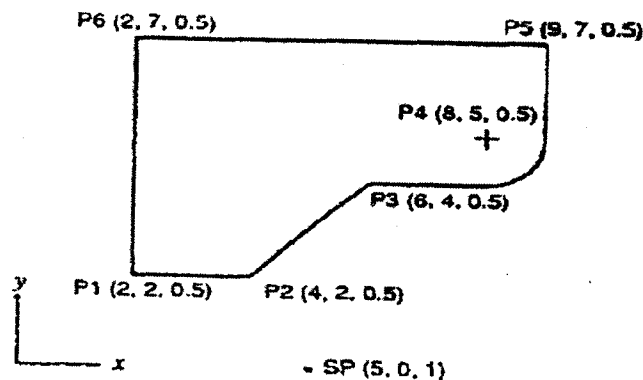
(P.T.O.)

- IV. (a) Differentiate between open and closed loop controls used in NC machines with neat sketches. (10)
- (b) Mention the G code formats for the following operation indicating tool used and spindle speed. (5)
- Linear interpolation
  - Circular interpolation (clockwise and anticlockwise)

OR

- V. Write short notes on the following (15)
- Incremental encoder
  - Absolute encoder
  - Linear position measuring transducers.

- VI. Write a detailed APT programme to a machine component given below. Assume P4 to be as the centre of the quarter circle of radius 1 unit. The thickness of the component is 0.50mm. SP is the start point of operation. (15)



OR

- VII. Briefly explain the design changes required in a CNC machine compared to a conventional machine? Give reason for each change. (15)
- VIII. (a) Discuss resolution, accuracy and repeatability of a robot. (5)
- (b) Discuss the various elements of an FMS. (10)
- OR
- IX. (a) Discuss any 3 sensors used in robots. (9)
- (b) Discuss the configuration of a SCARA robot. (6)

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